When Does Aggregating Explanations Work?

BreastMNIST Results

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This report consolidates the released BreastMNIST experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Full-matrix protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Result source snapshot: d6e4b9bbfb3c from Workshop-When-Does-Explanations.

Table 1: NAIVE results for `BreastMNIST` with `ViT-B/16` (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized. The test split contains $n = 156$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Best Individual	0.186	0.006	0.263	0.032	-0.051	0.013	-0.051	-0.051	0.006	-0.026	0.006	-0.077	0.000	-0.026	-0.026	-0.038	-0.038	-0.096	-0.051	-0.135
Simple Averaging	0.122	0.006	0.173	0.096	0.013	-0.026	0.051	0.000	0.000	-0.013	0.064	0.013	0.038	0.006	0.064	0.032	0.000	-0.064	0.026	-0.051
Borda	0.090	0.038	0.179	0.103	0.026	0.026	0.013	0.026	0.045	0.032	0.058	-0.006	0.026	0.019	-0.026	0.032	0.032	-0.064	0.019	-0.051
Kemeny–Young	0.090	0.026	0.154	0.115	0.038	-0.013	0.051	0.026	0.038	-0.019	0.051	-0.006	0.026	0.006	0.026	0.032	0.019	-0.096	0.019	-0.058
RRF	0.083	0.019	0.147	0.083	0.026	-0.019	0.038	-0.032	0.071	-0.006	0.083	-0.045	0.032	-0.006	0.019	0.006	0.045	-0.077	0.045	-0.064
Schulze	0.109	0.032	0.160	0.096	0.000	0.000	0.026	-0.013	0.019	0.026	0.045	-0.026	0.038	0.026	0.038	0.013	0.019	-0.090	0.032	-0.077

Table 2: IND results for `BreastMNIST` with `ViT-B/16` (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized. The test split contains $n = 156$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Best Individual	baseline	0.024	0.190	0.056	0.254	0.036	0.013	0.021	0.000	0.002	0.058	0.013	0.002	0.002	0.026	0.000	0.000	0.032	0.002	0.030	0.006
Simple Averaging	matched NAIVE	0.017	0.216	0.056	0.284	0.006	0.058	0.011	0.053	0.000	0.083	0.000	0.028	0.021	0.058	0.000	0.049	0.026	0.013	0.034	0.013
Simple Averaging	IND	0.019	0.199	0.058	0.288	0.019	0.064	0.058	0.103	-0.006	0.026	0.045	0.051	0.006	-0.006	0.006	0.032	0.045	0.013	0.045	0.051
Borda	matched NAIVE	0.026	0.252	0.051	0.321	0.013	0.058	0.043	0.062	-0.002	0.077	0.011	0.038	0.000	0.045	0.000	0.002	0.006	0.026	0.024	0.017
Borda	IND	0.032	0.218	0.071	0.256	0.038	0.064	0.064	0.077	0.000	0.090	0.026	0.026	0.013	0.038	0.013	0.000	0.006	0.058	0.032	0.045
Kemeny–Young	matched NAIVE	0.026	0.276	0.056	0.344	0.030	0.088	0.034	0.088	0.002	0.115	0.006	0.081	0.000	0.060	-0.004	0.030	0.011	0.006	0.024	0.002
Kemeny–Young	IND	0.058	0.212	0.083	0.263	0.006	0.083	0.071	0.109	-0.026	0.071	0.038	0.045	-0.013	0.045	0.026	0.019	0.013	0.071	0.038	0.058
RRF	matched NAIVE	0.030	0.274	0.056	0.368	0.019	0.107	0.036	0.124	0.009	0.088	0.021	0.066	0.000	0.060	-0.004	0.043	-0.002	0.043	0.019	0.073
RRF	IND	0.032	0.167	0.083	0.205	0.026	0.032	0.077	0.019	-0.013	0.038	0.051	-0.026	0.026	-0.019	0.026	-0.045	0.026	0.006	0.026	0.019
Schulze	matched NAIVE	0.030	0.259	0.060	0.340	0.017	0.077	0.034	0.085	-0.004	0.077	0.009	0.051	0.000	0.043	-0.009	0.013	0.017	0.017	0.026	0.026
Schulze	IND	0.038	0.192	0.077	0.256	0.032	0.064	0.045	0.090	-0.013	0.051	0.013	0.051	-0.013	0.019	0.000	0.006	0.006	0.019	0.032	0.006

Table 3: NOISE-S results for `BreastMNIST` with `ViT-B/16` (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	3	0.147	-0.006	0.224	0.058	0.032	-0.013	0.032	0.000	0.032	0.000	0.045	-0.026	0.064	-0.013	0.077	-0.013	0.026	-0.064	0.013	-0.077
Borda	3	0.154	0.013	0.244	0.064	0.032	0.000	0.058	0.000	0.013	0.000	0.000	-0.013	0.096	0.038	0.058	0.013	0.045	-0.103	0.006	-0.115
Kemeny–Young	3	0.179	0.000	0.231	0.026	-0.006	0.006	0.006	-0.058	-0.013	0.013	0.013	-0.051	0.013	-0.019	0.013	-0.045	-0.013	-0.096	0.000	-0.109
RRF	3	0.160	0.006	0.250	0.083	0.019	0.006	0.019	0.019	0.006	-0.013	0.006	0.013	0.045	0.013	0.006	0.026	0.032	-0.115	0.006	-0.103
Schulze	3	0.147	-0.006	0.250	0.045	0.064	-0.006	0.064	-0.045	0.071	-0.013	0.045	-0.013	0.083	0.013	0.032	0.000	0.032	-0.103	-0.006	-0.115

Table 4: NOISE-K results for `BreastMNIST` with `ViT-B/16` (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	3	0.147	-0.006	0.224	0.058	0.032	-0.013	0.032	0.000	0.032	0.000	0.045	-0.026	0.064	-0.013	0.077	-0.013	0.026	-0.064	0.013	-0.077
Borda	3	0.154	0.013	0.244	0.064	0.032	0.000	0.058	0.000	0.013	0.000	0.000	-0.013	0.096	0.038	0.058	0.013	0.045	-0.103	0.006	-0.115
Kemeny–Young	3	0.179	0.000	0.231	0.026	-0.006	0.006	0.006	-0.058	-0.013	0.013	0.013	-0.051	0.013	-0.019	0.013	-0.045	-0.013	-0.096	0.000	-0.109
RRF	3	0.160	0.006	0.250	0.083	0.019	0.006	0.019	0.019	0.006	-0.013	0.006	0.013	0.045	0.013	0.006	0.026	0.032	-0.115	0.006	-0.103
Schulze	3	0.147	-0.006	0.250	0.045	0.064	-0.006	0.064	-0.045	0.071	-0.013	0.045	-0.013	0.083	0.013	0.032	0.000	0.032	-0.103	-0.006	-0.115